



Project report

Vaultique

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Section 1



Task distribution:

Task	Student names
Project organization	Safaa Ahmed Sobh
Project idea & justification	Safaa Ahmed Sobh, Layan Alshreef
Problem Statement	Safaa Ahmed Sobh, Nour Zehrawi
Scope of work	Safaa Ahmed Sobh
Timeline	Ethar Hatem Tamim



1. Project Idea & Justification

The Idea: Vaultique – The "Living" Legacy Archive

Vaultique is a hybrid Digital-Physical curation platform designed for the "Aesthetic Generation." It reimagines traditional storage as a **Digital Time Capsule**. Unlike standard storage apps that simply list what is in a box, Vaultique creates an interactive, high-design **Digital Flipbook** for every major life chapter.

Justification:

1. The "Sentimental Clutter" Paradox (Psychological Justification)

The primary driver for Vaultique is the psychological weight of physical items.

- **The Problem:** Most people keep "stuff" because of the memories attached, not the item itself. This leads to cluttered homes and stress.
- **The Vaultique Solution:** By creating a high-fidelity digital "twin" of a physical item (a photo of a graduation gown + a video of the ceremony), the user feels emotionally "safe" to put the physical item in off-site storage.
- **Impact:** It solves the hoarding instinct by ensuring that "storing it" doesn't mean "forgetting it."

2. Alignment with the "Aesthetic Archiving" Trend

There is a massive cultural shift (driven by TikTok and Instagram) toward **curating** life rather than just living it.

- **Trend Analysis:** Modern users are obsessed with "Monthly Dumps" and "Travel Books." They want their memories to look professional and organized.
- **The Gap:** Standard storage companies like wheeKeep or Sitebox focus on boxes and trucks. They are "ugly" and industrial.
- **Vaultique's Edge:** Vaultique justifies its existence by being the **first "Beautiful" storage interface**. It turns a warehouse inventory into a luxury digital coffee table book. It moves storage from the "Logistics" category into the "Lifestyle" category.



3. The Digital-Physical "Deep Link" (Technological Justification)

Standard photo apps (Google Photos/iCloud) are "digital graveyards"—photos go in and are rarely seen again.

- **The Innovation:** Vaultique uses **QR-Deep Linking**. This justifies the software as a utility tool. When a user stands in front of a physical box, they shouldn't have to guess what's inside.
- **Efficiency:** For milestones like **Graduation** or **Weddings**, users can instantly recall the specific video clip of that moment by interacting with the physical box. This "Phygital" (Physical + Digital) connection is a unique selling point that justifies a premium subscription model.

4. Long-term Legacy & Intergenerational Value

We are the first generation to leave behind a massive digital footprint, yet it is highly disorganized.

- Vaultique serves as a **curated legacy tool**. Instead of leaving your children 50 unlabeled boxes and a hard drive with 10,000 random photos, you leave them a "Vaultique Account" organized into "Life Chapters" (e.g., *Our First Home*, *The Europe Trip*, *Sarah's Graduation*).
- **Market Need:** There is a growing market for **Legacy Tech**—tools that help people organize their life's work for the next generation. Vaultique provides a structured, beautiful way to do this.

5. Urbanization and the "Space Economy."

As more people move into smaller apartments in cities like Riyadh or New York, the "Price per Square Foot" makes keeping sentimental items at home expensive.

- **Economic Justification:** It is cheaper to pay for a Vaultique-managed storage unit than to rent a larger apartment just to hold "memories."
- **Utility:** By providing software that makes remote storage feel "accessible and visible," users are more likely to utilize professional storage services, as seen in the growth of companies like Masaha and WheeKeep.



2. Problem Statement

The modern consumer experiences a phenomenon known as "**Memory Fragmentation**." This problem consists of three core pain points:

- **The "Digital Graveyard":** While digital photos are easily captured, they are often buried in massive, unorganized camera rolls. Without curation, these photos lose their narrative power and are rarely revisited.
- **Sentimental Clutter & Space Scarcity:** Physical keepsakes (graduation gowns, baby shoes, wedding decor, travel souvenirs) occupy valuable living space. Many individuals struggle with "sentimental hoarding"—the inability to discard items because of the memories they trigger.
- **The Context Gap:** Current storage solutions, such as those provided by wheeKeep or Sitebox, are strictly logistical. There is no existing tool that links the *physical item* in a box to the *digital story* of the day it was acquired.

Vaultique solves this by providing a unified platform where physical items are organized into "**Event Chapters**" that mirror a premium digital photobook, ensuring that "storing it" does not mean "forgetting it".



3. Scope of Work

A. User Interface & Experience (Aesthetic Curation)

- **Milestone Templates:** Custom-designed digital "Chapter" layouts for **Graduations, Births, Weddings, and Travel**.
- **Interactive Scrapbook Engine:** A front-end interface that allows users to drag-and-drop photos, videos, and text into an automated, aesthetic "book" format.
- **Life-Map View:** A geographical dashboard for travel milestones, allowing users to "click" on a country to open that specific digital book.

B. Physical-Digital Integration (The "Phygital" Link)

- **QR Code Ecosystem:** A module to generate unique, high-resolution QR labels for physical storage boxes.
- **Deep-Link Scanner:** An in-app camera tool that, when pointed at a box's QR code, instantly navigates the user to the corresponding digital chapter and item inventory.
- **Itemized Souvenir Logging:** The ability to photograph individual items *within* a box and tag them with "Memory Clips" (voice notes or short videos)

C. Data & Media Management

- **High-Fidelity Hosting:** Secure cloud storage for 4K video and high-res images to ensure memories do not degrade over time.
- **Metadata Harvesting:** Automatic organization of photos by GPS location and timestamp to suggest "Milestone Chapters" to the user.
- **Family Collaboration:** A "Contributor" mode where family members (e.g., grandparents or spouses) can upload their own photos to a shared milestone book.

D. Out of Scope (Exclusions)

- **Physical Moving Services:** Vaultique provides the *software* for management; it will not own a fleet of moving trucks.
- **E-Commerce/Souvenir Sales:** The app will not sell items; it is strictly an archive and management tool.
- **Printing Services:** Physical printing of books is excluded from the initial MVP (Minimum Viable Product).



4. Timeline

Phase 1: Discovery & Aesthetic Architecture (Weeks 1–2)

Focus: Defining the visual identity and the logic of the "Milestone Chapters."

- **Week 1 (March 3 – March 7):** Market analysis of photobook trends (e.g., Journi or Pixory) and definition of "Milestone" templates (Graduation, Baby, Wedding, Travel).
- **Week 2 (March 10 – March 14): UI/UX Prototyping.** Designing the "Flipbook" animation and the "Map-View" dashboard. Finalizing the Database Schema (SQL/NoSQL) to link digital media to physical box IDs.
- **Milestone 1 (March 14): Design Freeze & System Architecture Document.**

Phase 2: Core Development & Cloud Setup (Weeks 3–5)

Focus: Building the engine that handles the data.

- **Week 3 (March 17 – March 21): Backend Foundation.** Setting up secure cloud hosting (AWS/Google Cloud) for high-resolution 4K video and photo storage.
- **Week 4 (March 24 – March 28): The "Scrapbook" Engine.** Developing the algorithm that uses photo metadata (GPS and timestamps) to automatically group images into "Chapters."
- **Week 5 (March 31 – April 4): User Identity Module.** Implementing secure authentication (OAuth) and the "Family Collaboration" invitation system.
- **Milestone 2 (April 4): Functional Backend & Alpha "Chapter" Generator.**

Phase 3: The "Phygital" Integration (Weeks 6–8)

Focus: Connecting the physical storage box to the digital app.

- **Week 6 (April 7 – April 11): QR Code Ecosystem.** Building the generator that creates unique, high-resolution QR labels for physical boxes as seen in modern Portable Storage Guides.
- **Week 7 (April 14 – April 18): The Deep-Link Scanner.** Developing the in-app camera tool that "reads" the box and instantly triggers the correct digital flipbook.
- **Week 8 (April 21 – April 25): Multimedia Sync.** Enabling "Memory Clips" — allowing users to record audio/video notes and "attach" them to specific items within the digital box view.
- **Milestone 3 (April 25): Beta "Phygital" Sync (Physical Box to App Connection).**



Phase 4: Advanced Features & Aesthetic Polishing (Weeks 9–10)

Focus: Perfecting the "Travel Book" vibe and map integration.

- **Week 9 (April 28 – May 2): Interactive Map API.** Integrating Mapbox or Google Maps to visualize "Travel Milestones" geographically.
- **Week 10 (May 5 – May 9): Front-end Refinement.** Adding "micro-interactions" (page-flip sounds, smooth transitions, and high-end typography) to mimic a luxury coffee table book.
- **Milestone 4 (May 9): Feature-Complete MVP (Minimum Viable Product).**

Phase 5: Quality Assurance & Launch Preparation (Weeks 11–12)

Focus: Stress-testing and final delivery.

- **Week 11 (May 12 – May 16): UAT (User Acceptance Testing).** Testing with "real-world" milestone data (e.g., uploading 500 graduation photos at once) to ensure zero lag. Bug fixing and security audits.
- **Week 12 (May 19 – May 23): Final Deployment.** Preparing the PDF/Word documentation and the final presentation. App deployment to testing environments (TestFlight/App Store Connect).
- **Milestone 5 (May 23): Project Handover & Live Demonstration.**

5. Summary of Key Milestones

Phase	Duration	Start Date	End Date	Key Milestone
1. Design	2 weeks	March 3	March 14	UI Prototype Ready
2. Backend	3 weeks	March 17	April 4	Database & AI Logic Live
3. Phygital	3 weeks	April 7	April 25	QR Scanner Functional
4. Polishing	2 weeks	April 28	May 9	Interactive Map Integrated
5. Testing	2 weeks	May 12	May 23	Final Project Handover



6. The Project Team Roles

1. The Strategy & Logic Layer (Weeks 1–3)

These roles start first to ensure the project has a goal and a structure.

- **Project Manager (PM):** Defines the "Milestone Chapters" and handles market analysis. They ensure the team doesn't build features that nobody wants.
- **System Architect / Backend Developer:** They design the **Database Schema** (SQL/NoSQL) and decide how the "Phygital" box ID will talk to the digital media.
- **UI/UX Designer:** Creates the "Aesthetic Architecture." They decide how the flipbook looks and feels before a developer writes the code.

2. The Foundation & Infrastructure (Weeks 3–5)

Once the plan is set, the "heavy lifting" begins on the servers.

- **DevOps Engineer:** Sets up the Cloud hosting (AWS/Google Cloud). Without them, the developers have nowhere to store those 4K "Memory Clips."
- **Backend Developer:** Build the "Scrapbook Engine." They write the algorithms that sort photos by GPS and timestamps.

3. The Interactive Layer (Weeks 6–10)

Now that the data exists, it needs to be visible and interactive.

- **Full-Stack / Mobile Developer:** They build the **QR Code Scanner**. This is the bridge between the physical box and the digital app.
- **Frontend Developer:** They integrate the Map APIs and add the "Luxury" polish—the smooth page-flips, sounds, and high-end typography.

4. The Integrity & Delivery Layer (Weeks 11–12)

The final step is making sure nothing breaks under pressure.

- **QA Engineer (Quality Assurance):** They try to "break" the app by uploading 500 photos at once or scanning blurry QR codes to ensure a zero-lag experience.
- **Security Lead:** Performs the final audit on the OAuth user authentication to make sure "Family Collaboration" invites are secure.



7. Summary of Project Team Roles

Phase	Core Technical Focus	Person in charge
1. Design	UI/UX & Architecture	Project Manager System Architect / Backend Developer UI/UX Designer
2. Backend	Data Engineering	DevOps Engineer Backend Developer
3. Phygital	IoT & Integration	Full-Stack / Mobile Developer Frontend Developer
4. Polishing	Front-end Engineering	Front-end Engineer
5. Testing	Quality Assurance	QA Engineer Security Lead



Section 2



Task distribution:

Task	Student names
Project organization	Safaa Ahmed Sobh
Functional requirements analysis	Nour Zehrawi
Non-Functional requirements analysis	Layan Alshreef
User analysis	Safaa Ahmed Sobh
Risk assessment and mitigation plan	Ethar Hatem Tamim



1. Requirements analysis:

Functional:

Notation	Description
1. User Story Management & Customization (Requirements Elicitation)	User Stories: The system allows users to define their requirements as "User Stories" ("As a user, I want to create a graduation chapter to preserve my university memories"). Milestone Templates: The system shall provide predefined templates for common life events (Graduation, Wedding, Travel) to simplify the specification process for end-users. Collaborative Editing: The system shall support a "Contributor Mode" where multiple stakeholders (family members) can collaborate on the same digital chapter.
2. Digital Content Curation (System Services)	Multimedia Upload: The system shall provide an interface for users to upload and organize 4K videos, high-resolution photos, and text into a "Digital Flipbook" format. Automated Metadata Harvesting: The system shall automatically group media into "Chapters" based on GPS location and timestamps (Metadata). Geographical Visualization: The system shall integrate a "Life-Map View" (using Google Maps API) to visualize travel milestones on a map.
3. "Phyigital" Integration (System Interaction)	QR Code Generation: The system shall generate unique, high-resolution QR labels for every physical storage box registered in the system. Deep-Link Scanning: The system shall include an in-app scanner that, when pointed at a physical QR code, navigates the user to the specific digital chapter and item inventory. Memory Clips: The system shall allow users to record and attach audio or video "Memory Clips" to specific physical items itemized within the digital box view.
4. System Evolution & Maintenance	Search Functionality: The system should provide a search service allowing users to find specific items or chapters across their entire archive. Refactoring Support: To maintain simplicity and code quality, the system should be designed to support continuous refactoring of the digital archive structure. Security & Authentication: The system shall uniquely identify and authenticate every staff member or user using a secure identity module (OAuth).



Non-functional:

Notation	Description
Performance	The system shall provide a fast response time, where digital flipbooks and media content load within 2–3 seconds. It shall efficiently handle high-resolution media and support multiple users simultaneously.
Security	The system shall ensure data protection through encryption during storage and transmission. It shall implement secure authentication and prevent unauthorized access.
Usability	The system shall provide an intuitive, user-friendly interface with an aesthetic design, enabling easy navigation between chapters and content.
Availability	The system shall be available 24/7 with minimal downtime and ensure continuous access through cloud services.
Reliability	The system shall operate without data loss or corruption and include backup mechanisms with quick recovery from failures.
Scalability	The system shall support the increasing number of users and growing data efficiently to enable future expansion.
Compatibility	The system shall be compatible with mobile, tablet, and desktop platforms, supporting iOS, Android, and web browsers.
Maintainability	The system shall be easy to maintain and update, supporting future enhancements and continuous improvement.



2. User analysis:

User Personas:

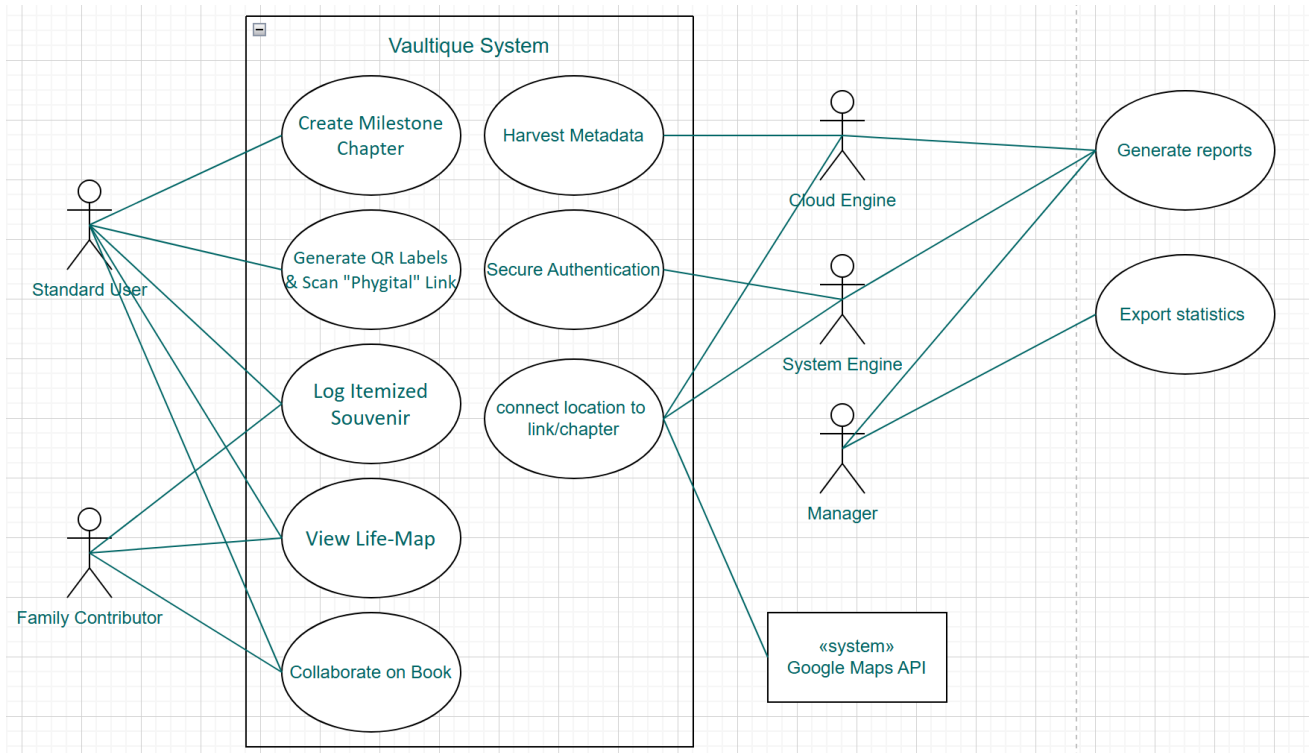
Persona Name	Role/Profile	Primary Need
The New Graduate	Recent university student.	Wants to store their graduation gown and cap physically but keep the ceremony video and photos easily accessible via a QR scan.
The Global Traveler	Active traveler who values "Monthly Dumps".	Needs a "Life-Map View" to organize souvenirs from different countries into phygital "Travel Books".
The Legacy Curator	A parent or grandparent looking to organize a "Living Legacy".	Wants to replace 50 unlabeled boxes with a curated "Vaultique Account" for the next generation.

Use Case:

Actor	Use Case (Goal)	Description / Details
Standard User	Create Milestone Chapter	Users drag-and-drop photos, videos, and text into automated, aesthetic "book" formats.
	Generate QR Labels	The system creates unique, high-resolution QR labels for physical storage boxes.
	Scan "Phygital" Link	Using the in-app camera to scan a box QR and instantly navigate to the digital flipbook.
	Log Itemized Souvenir	Photographing individual items and tagging them with "Memory Clips" (voice/video).
	View Life-Map	Using a geographical dashboard to click on a country and open travel-specific books.
Family Contributor	Collaborate on Book	Family members (grandparents/spouses) upload their own media to a shared milestone book.
System (Automated)	Harvest Metadata	Automatically organizing photos by GPS and timestamp to suggest "Milestone Chapters".
	Secure Authentication	Implementing secure OAuth for user identity and family invitations.



Diagram:





3. Risk assessment & mitigation plan:

Required Tools & Resources	mitigation Operations & Technical Solutions	Risk Level	Risk Scenario	Risk Category
Custom crates, shock/vibration sensors, hydraulic tie-down straps.	• Use custom-built wooden crates lined with shock-absorbing foam. • Install impact sensors on crates to alert the driver of rough handling. • Specialized "soft handling" training for the moving crew.	High	Damage to items (furniture, instruments) due to vibration, shocks, or improper loading.	Transit Risks
RFID tags & scanners, WMS software, high-res handheld scanners.	• Implement a Smart Warehouse Management System (WMS) for real-time tracking. • Attach RFID tags or QR codes to every item to log entry dates and exact shelf location. • Conduct monthly visual digital audits.	Medium	Loss of small items or mixing belongings with other customers' inventory.	Logistics Risks
IoT climate sensors, industrial dehumidifiers, thermal insulation.	• Install redundant HVAC systems to ensure a backup if one unit fails. • Deploy automated dehumidifiers triggered by moisture thresholds. • Use breathable thermal wraps that prevent moisture buildup.	Medium	Warping of wood, mold on paintings, or fabric decay due to high humidity/heat.	Environmental Risks
Digital pest monitors, vacuum-seal kits, and airtight storage containers.	• Store items on plastic pallets (elevated 20 cm) to avoid organic wood attraction. • Use vacuum-sealing technology for fabrics and sensitive upholstery. • Install digital traps that send instant alerts upon any activity.	Low	Wood-boring insects or rodents are damaging furniture, paper archives, or textiles.	Pest Risks



Required Tools & Resources	mitigation Operations & Technical Solutions	Risk Level	Risk Scenario	Risk Category
Gas fire suppressors, laser smoke detectors, fire-rated doors.	• Install Clean Agent (FM-2 suppression)—it's without water damage. • Use 4-hour fire-rated compartmentalization walls to contain outbreaks. • Deploy high-sensitivity laser smoke detectors (VESDA).	Critical	Ignition of flammable stored materials (wood, paper, and fabrics) causes total loss.	Fire Risks
Biometric locks, 4K AI cameras, and silent alarm monitoring systems.	• Implement biometric access (Face ID/fingerprint) for high-security zones. • 4K CCTV with night vision and AI-powered facial recognition. • Provide a live feed option for premium clients to view their items remotely.	High	Internal theft, external break-ins, or intentional vandalism of high-value goods.	Security Risks
360° cameras , digital contract platforms , automated payment gateways.	• 360-degree high-definition photography of every item upon receipt as a baseline. • Offer "agreed value" insurance policies rather than standard weight-based coverage. • Automated recurring billing with legally binding digital notification trails.	Medium	Disputes over item condition at pickup or non-payment of storage fees.	Legal Risks



Section 3



Task distribution:

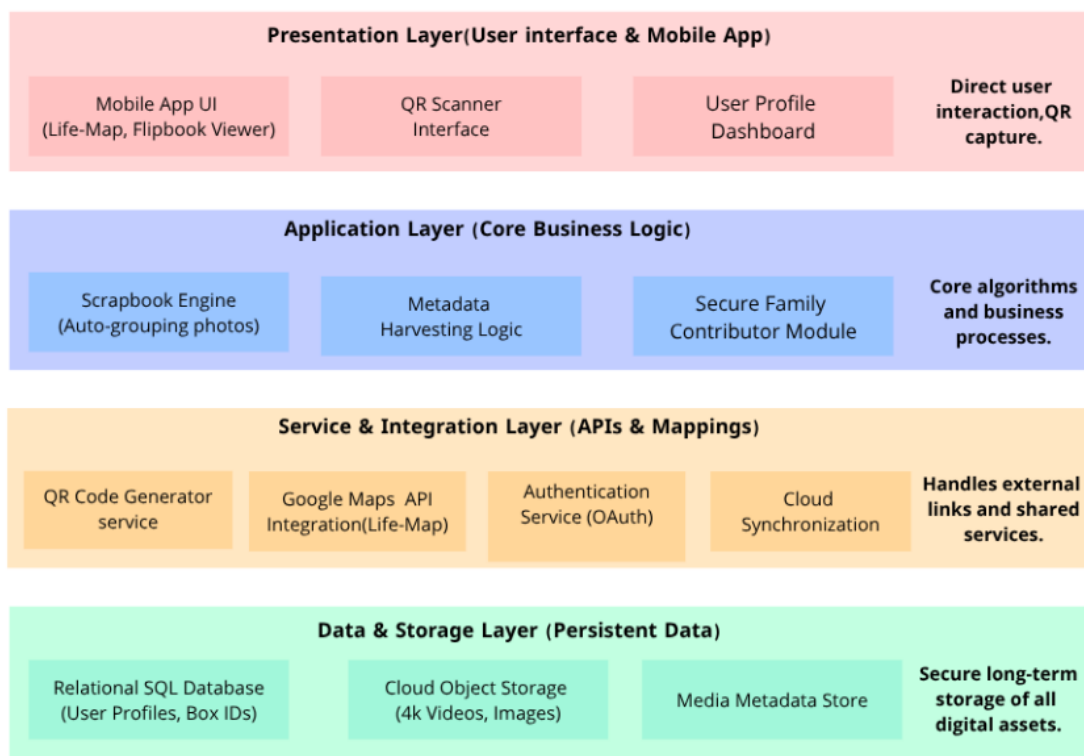
Task	Student names
Project organization	Safaa Ahmed Sobh
Software architecture overview	Nour Zehrawi
UI wireframes/mockups	Safaa Ahmed Sobh, Ethar Hatem Tamim
UML diagrams with explanations	Layan Alshreef



1. Software Architecture Overview:

Decision Topic	Choice	Rationale
System Distribution	Client-Server Architecture	To allow mobile users to access cloud-hosted memories remotely from any location
Architectural Style	Layered Architecture	To support incremental changes and separate the user interface from the complex data processing.
Control Strategy	Centralized Control	A central server manages the synchronization between physical box IDs and digital Life Chapters.

1.1 Architectural Patterns (Layered Model)





1.2 System Decomposition (Component Responsibilities)

Component	Responsibility
QR Linker	Generates and decodes unique IDs to link physical souvenirs to digital events.
Event Organizer	Automatically groups photos/videos by location and date to create "Milestone Chapters".
Security Module	Implements OAuth for secure family-shared access to private archives.

1.3 System Communication Flow

The system operates as follows:

1. The user interacts with the application through the Presentation Layer.
2. The request is sent to the Application Layer (API).
3. The request is processed in the Business Logic Layer.
4. Data is stored or retrieved from the Data Layer.
5. The response is sent back to the user through the same layers.

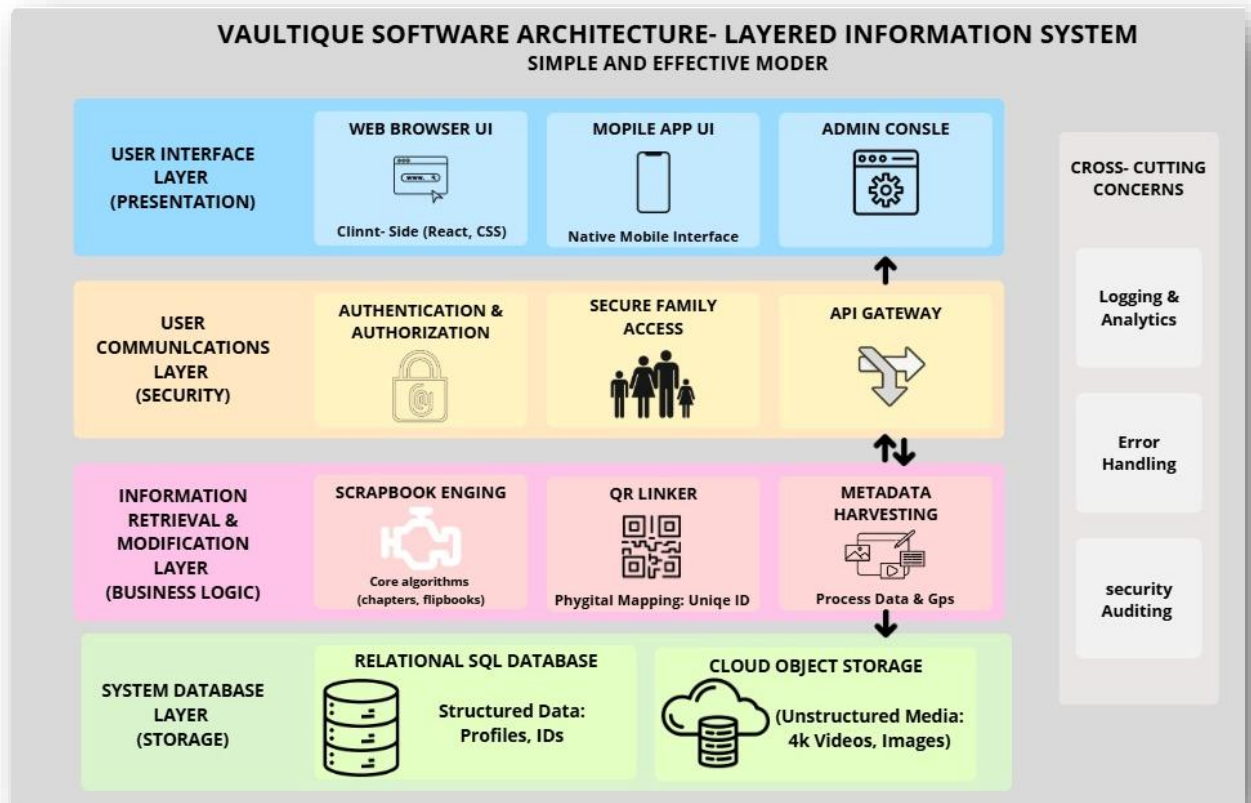
1.4 Architectural Attributes Mapping

The chosen architecture optimizes the following system characteristics:

- **Performance:** By using localized processing in the Application Layer, we ensure that digital flipbooks load within 2-3 seconds.
- **Security:** Critical data is isolated within the inner layers of the architecture to prevent unauthorized access.
- **Maintainability:** The independent nature of the sub-systems allows for future updates (e.g., adding VR views) without re-architecting the entire system.

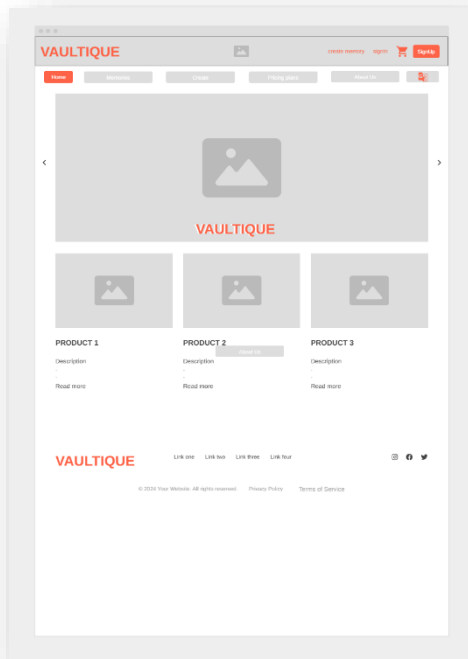


1.5 Architecture Diagram





2.1 UI Wireframes Prototype:



2.1.1 Home page:

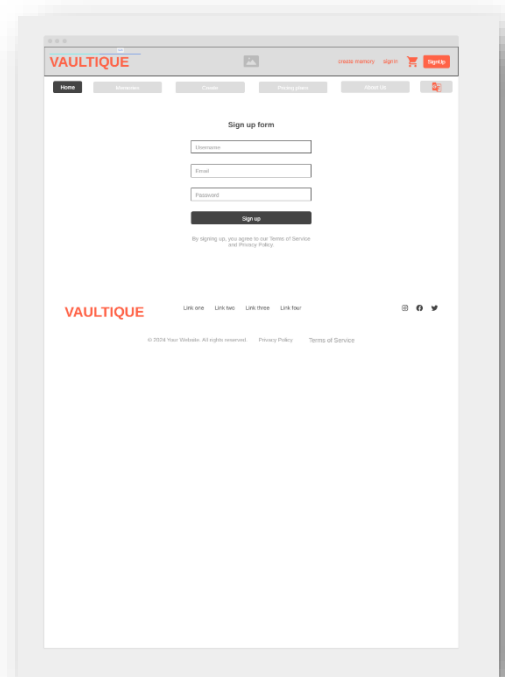
This page serves as the **Home Page** for the platform, acting as the primary entry point and navigation hub for users. Its main functions are:

- **User Onboarding:** It uses the top-right "Sign Up" and "Sign In" buttons to manage user accounts.
- **Feature Navigation:** The menu bar directs users to the core tools, such as viewing their saved "Memories," creating new entries, or checking "Pricing plans."
- **Promotion & Discovery:** The central carousel highlights key announcements or brand identity, while the three product cards below introduce specific services or features available to the user.

2.1.2 Sign Up Page:

Functions as the primary portal for user registration and lead generation. Here is a breakdown of what this page does:

- **User Data Collection:** The central form is designed to capture essential user credentials—**Username, Email, and Password**—to create a new account in the system.
- **Legal Compliance:** By including the "Terms of Service and Privacy Policy" disclaimer below the button, the page ensures the user provides **informed consent** before joining the platform.
- **Navigation Consistency:** The header and footer remain identical to the home page, allowing users to exit the sign-up flow or return to the "Home" or "About Us" sections easily if they aren't ready to commit.

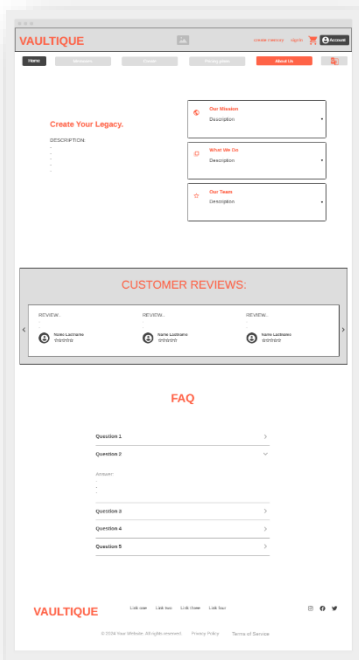
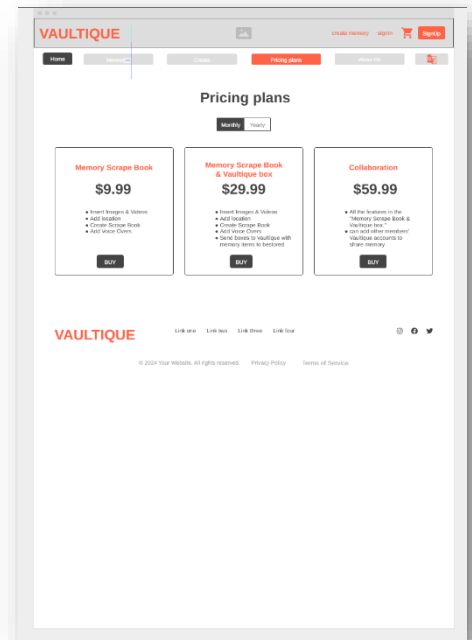




2.1.3 Pricing Plans Page:

Functions as the platform's **monetization and selection hub**. Its main roles are:

- **Service Tier Comparison:** It presents three distinct subscription levels—**Memory Scrape Book**, **Vaultique Box**, and **Collaboration**—allowing users to compare features like voice-overs, physical storage, and team sharing.
- **Revenue Conversion:** Through the "BUY" buttons and the **Monthly/Yearly toggle**, the page facilitates the transition from a casual visitor to a paying customer by offering clear price points.
- **Feature Education:** By listing bullet points under each plan, it informs users about the specific capabilities they are purchasing, such as the ability to "Send boxes to Vaultique" for physical restoration.



2.1.4 About Us Page:

Acts as the platform's **credibility and information center**. Its main functions are:

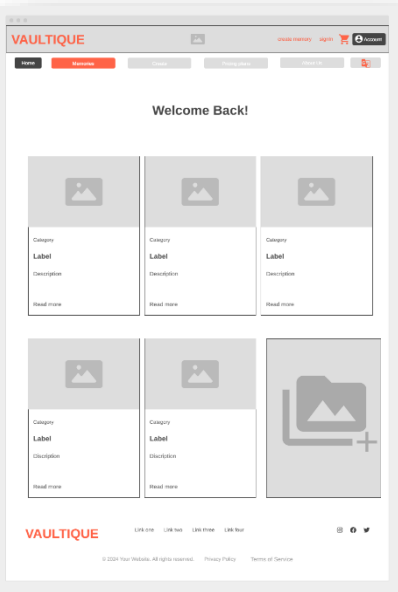
- **Brand Identity & Purpose:** It defines the company's values through sections like "**Our Mission**," "**What We Do**," and "**Our Team**," helping to build a narrative around "creating your legacy."
- **Social Proof:** The **Customer Reviews** carousel provides testimonials and ratings to build trust and demonstrate the platform's reliability to potential users.
- **User Support:** The **FAQ (Frequently Asked Questions)** section uses an accordion-style layout to proactively address common concerns and reduce the need for direct customer support inquiries.



2.1.5 Memories Dashboard:

Functions as the user's **personal digital library**. Its main roles are:

- **Content Management:** It displays a grid of the user's saved entries with "Category" and "Label", allowing them to browse and revisit their stored memories.
- **Quick Access:** The "Read more" links provide a path to view detailed information for each specific memory or archived item.
- **More files:** The large icon with a "+" sign in the bottom right acts as a **folder** for users to access more memory entries.
- **Personalization:** The "Welcome Back!" heading and the presence of an "Account" button indicate this is a **logged-in view**, tailored specifically to the individual user's profile.

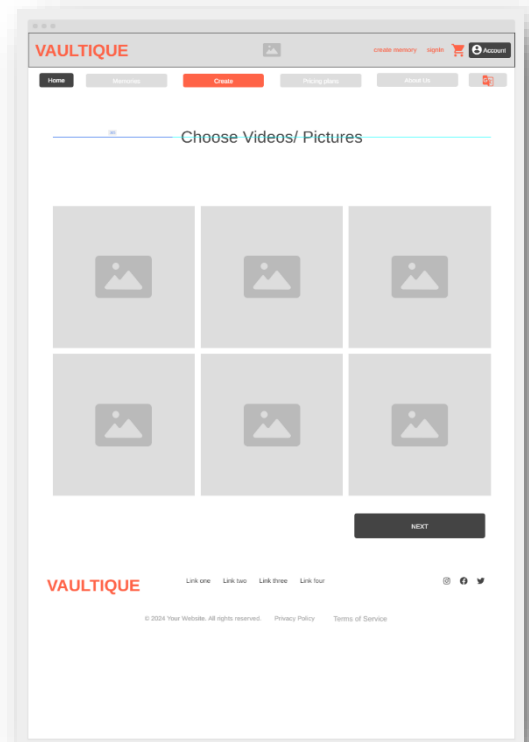


2.1.6 Create Memories UI:

Content Selection Page:

functions as the first step in the **creation workflow**. Its main roles are:

- **Media Selection:** It provides a grid interface for users to **browse and choose** specific videos or pictures from their device or cloud storage to be added to a new "Memory."
- **Workflow Progression:** The "NEXT" button serves as the primary action to move the user to the next stage of the creation process, such as adding descriptions or locations.
- **Visual Organization:** By using a placeholder grid, the page allows users to **preview and organize** multiple files at once before final submission.

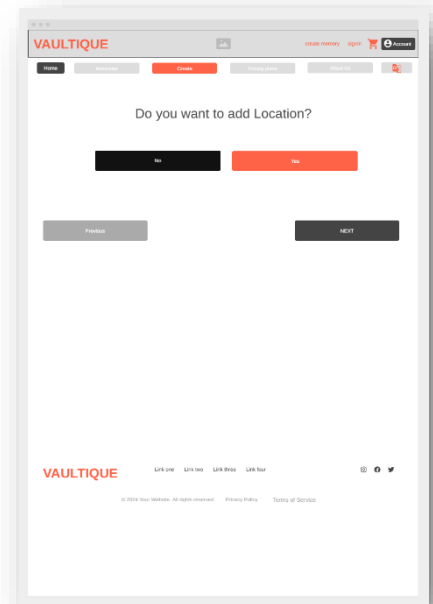




Metadata/Location - Step 1: confirmation

Acts as a **contextual refinement stage** in the creation process. Its main functions are:

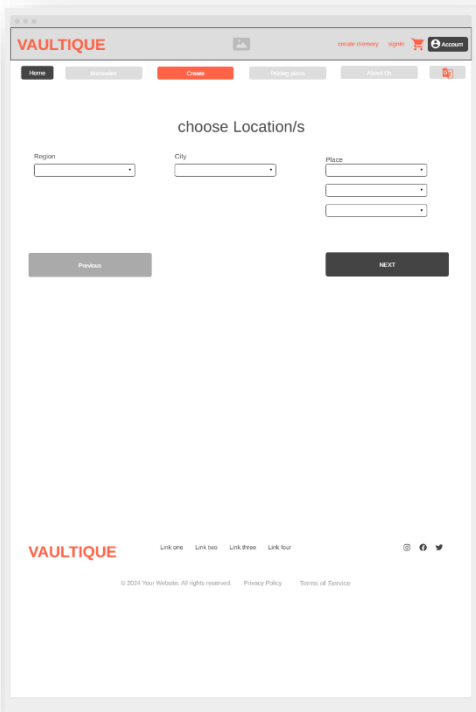
- **User Decision Point:** It uses a simple binary choice ("Yes" or "No") to determine if geographical data should be attached to the memory, allowing for a personalized and organized gallery.
- **Workflow Navigation:** The "Previous" and "NEXT" buttons ensure the user can move fluidly through the multi-step creation wizard without losing their progress.
- **Data Enrichment:** By prompting for a location, the page helps build a more detailed "story" for the digital vault, which later assists in filtering or searching memories by place.



Location Input Page - Step 2: choosing

Functions as the **data entry stage** for geographical tagging. Its main roles are:

- **Categorized Data Entry:** It uses hierarchical dropdown menus (**Region, City, and Place**) to help users precisely tag where their memory took place.
- **Detailed Mapping:** By providing multiple "Place" fields, it allows users to be as specific as possible, which improves future search and filtering capabilities within the digital vault.

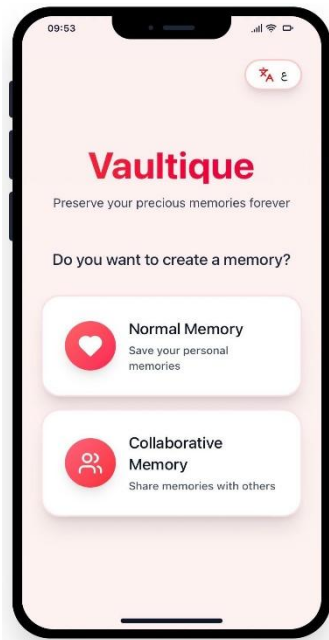




2.2 UI Mockup Prototype:

2.2.1 Mobile Create Memory Screen:

Acts as a **selection gateway** for starting a new entry. Its main functions are:

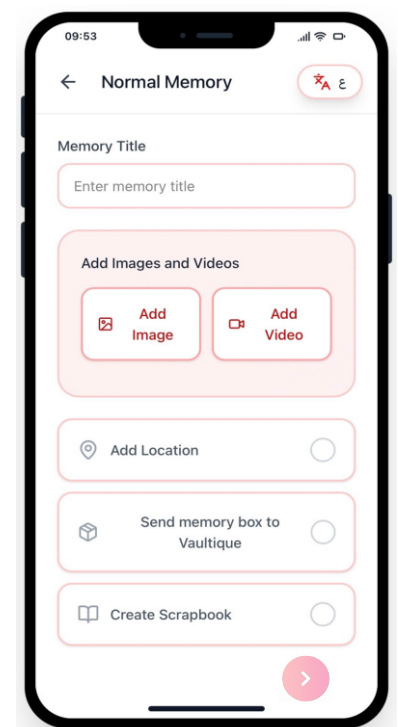


- **Mode Selection:** It offers two distinct paths for the user: a "**Normal Memory**" for personal use and a "**Collaborative Memory**" for shared experiences with others.
- **User Onboarding:** By asking the question "Do you want to create a memory?", it simplifies the decision-making process for the user immediately after landing on the creation tab.
- **Accessibility & Localization:** The language toggle button at the top right allows for quick switching (likely between English and Arabic), ensuring the app is accessible to a broader user base.
- **Mobile-First Design:** The large, touch-friendly cards and clear iconography are optimized for easy interaction on a smartphone screen.

2.2.2 Individual Memory Creation Screen:

serves as the primary data input interface for a single-user project. Its main roles are:

- **Content Identification:** It provides a text field for the user to assign a "**Memory Title**," ensuring the entry is easily searchable in their digital vault later.
- **Media Integration:** The dedicated "Add Image" and "Add Video" buttons act as **direct upload triggers**, allowing users to attach digital media to their entry.
- **Service Configuration:** It features a checklist of optional "Phygital" services, such as **Add Location**, **Send memory box to Vaultique**, and **Create Scrapbook**, enabling users to customize their order.
- **Progress Control:** The floating action button (the pink arrow) serves as the **submission or "Next" trigger**, advancing the user to the final confirmation or payment stage.

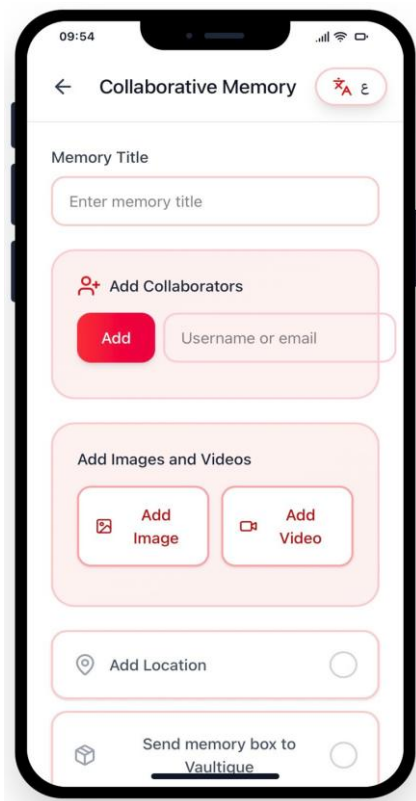
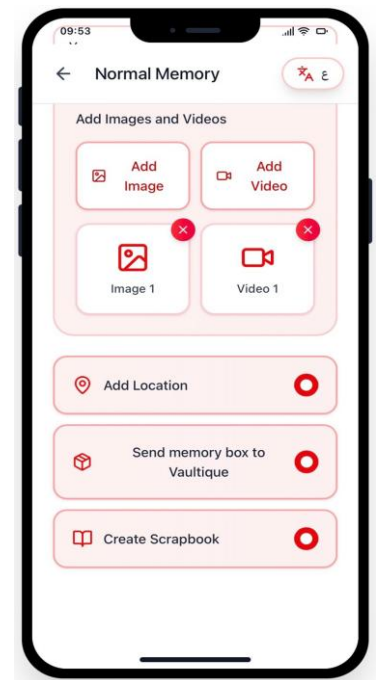




2.2.3 Content Review and Selection Screen:

Functions as the **finalization stage** for a specific memory entry. Its main roles are:

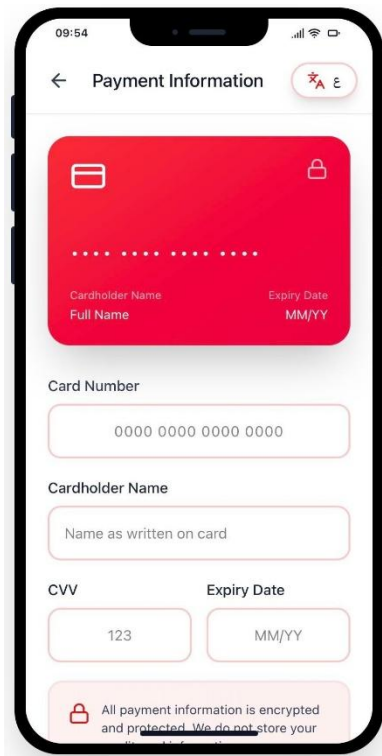
- **Media Management:** It provides a preview of uploaded files (like **Image 1** and **Video 1**) with "X" delete icons, allowing users to **edit or remove** content before proceeding.
- **Service Selection Confirmation:** The red circular indicators show that the user has successfully **toggled on** additional features like location tagging, physical box delivery, and scrapbook creation.
- **Visual Feedback:** By displaying both the upload buttons and the successfully added files in one view, it gives the user **clear confirmation** of their progress within the creation workflow.



2.2.4 Collaborative Memory Creation Screen:

Functions as the **multi-user project setup stage**. Its main roles are:

- **Shared Project Access:** It includes a dedicated "**Add Collaborators**" section, allowing the user to invite others to the memory via username or email.
- **Collective Content Input:** Similar to the individual mode, it provides a title field and buttons to **add images and videos**, but in a format designed for group contributions.
- **Joint Service Customization:** It offers the same "**Phyigital**" options—like location tagging and physical box delivery—so the group can decide which services to apply to their shared project.
- **User Management:** By integrating the collaborator field directly into the creation flow, the page ensures that **permissions and access** are established before the memory is finalized.



2.2.5 Payment Information Screen:

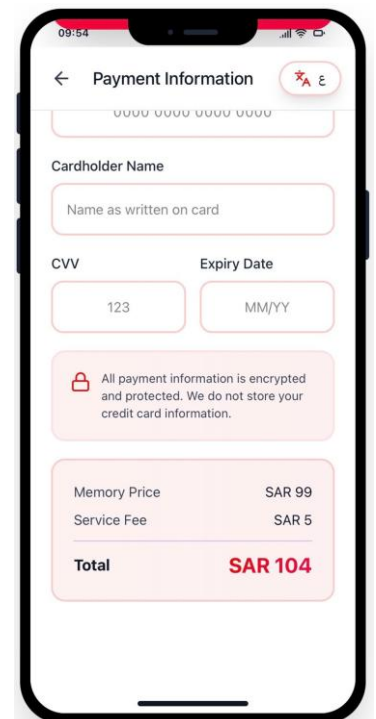
Functions as the **transaction and checkout stage** of the mobile app. Its main roles are:

- **Financial Data Collection:** It provides secure input fields for the user's **Card Number, Cardholder Name, CVV, and Expiry Date** to process service payments.
- **Visual Validation:** The dynamic card graphic at the top serves as a **live preview**, reflecting the user's input to help prevent entry errors before submission.
- **Internationalization:** Similar to other screens, it maintains the **language toggle** to ensure the checkout process is clear for users in different regions.

2.2.6 Order Summary and Payment Confirmation view:

Functions as the **final review stage** before a user completes their purchase. Its main roles are:

- **Financial Transparency:** It provides a detailed **price breakdown**, showing the base "Memory Price," a "Service Fee," and a prominent bold **Total**. Hence, the user knows exactly what they are being charged.
- **Security Assurance:** The screen includes a **security notice** with a lock icon, reinforcing the message that payment data is encrypted and not stored, which helps reduce checkout abandonment.





3.1 UML Class Diagram Explanation

3.1.1-Overview

The UML class diagram represents the structure of the Vaultique system. It illustrates how the main system components are connected, including the digital side (Chapter, Media, and MemoryClip) and the physical side (Box with QR code integration). The purpose of the system is to organize memories into structured “Life Chapters” instead of leaving them scattered and unorganized.

3.1.2-User Class

Purpose:

Represents the user of the system and manages account-related activities.

Attributes:

- userID : int
- name : String
- email : String
- password : String

Methods:

- login() : boolean → used for user authentication.
- createChapter() : void → allows the user to create a new memory chapter such as graduation, travel, or wedding memories.

Relationship:

There is a one-to-many (1..*) relationship between User and Chapter because one user can create multiple chapters.



3.1.3-Chapter Class

Purpose:

Represents the main organizational unit where memories are stored and categorized.

Attributes:

- chapterID : int
- title : String
- date : Date
- location : String

Methods:

- addMedia() : void → used to add photos or videos to the chapter.
- editChapter() : void → used to update or modify chapter information.

Relationship:

Each Chapter belongs to one User and can contain multiple Media files and MemoryClips.

3.1.4-Box Class

Purpose:

Represents the physical storage box connected to the digital system.

Attributes:

- boxID : int
- qrCode : String

Methods:

- generateQR() : String → generates a unique QR code for the box.

Relationship:

There is a one-to-one (1..1) relationship between Box and Chapter because each box is linked to one specific chapter.



3.1.5-Media and MemoryClip Classes

Media Class

Purpose:

Stores uploaded files such as images and videos.

Attributes:

- mediaID : int
- type : String
- file : File

Methods:

- upload() : void → used to upload media files.
- delete() : void → used to remove media files.

Relationship:

Each Media object belongs to one Chapter, while one Chapter can contain multiple Media files (1..*).

MemoryClip Class

Purpose:

Stores short audio or video clips related to specific memories or items.

Attributes:

- clipID : int
- type : String

Methods:

- recordClip() : void → used to record and attach clips.

Relationship:

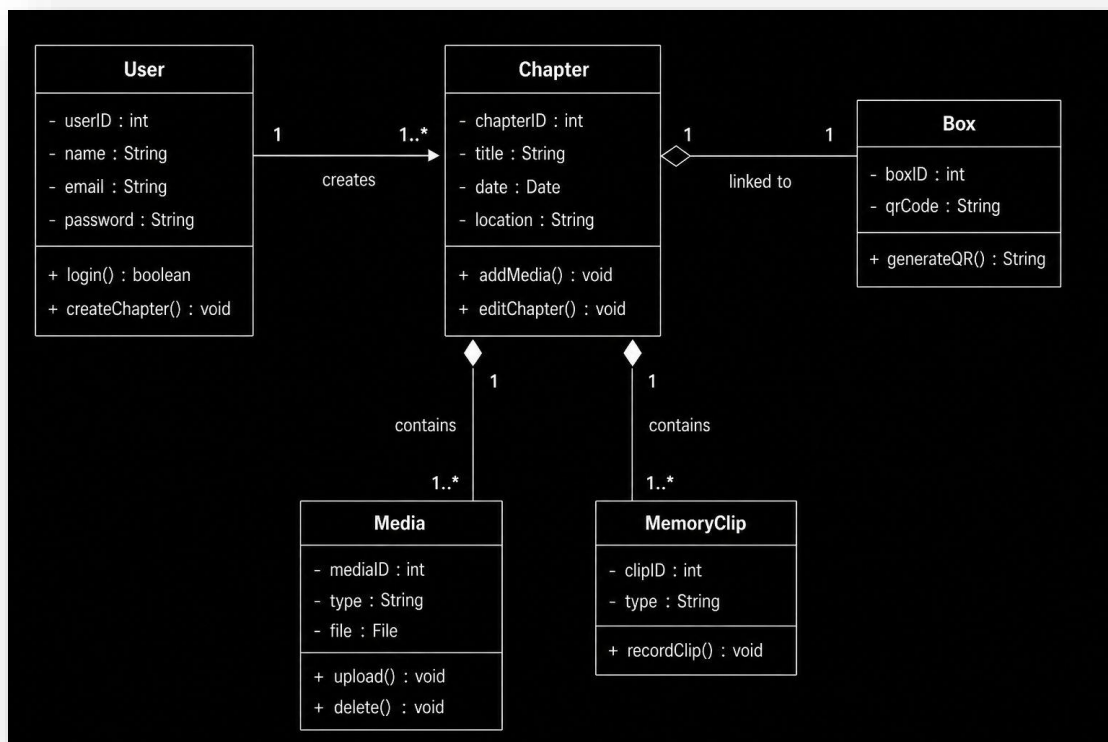
Each MemoryClip belongs to one Chapter, while one Chapter can contain multiple MemoryClips (1..*).



3.1.6-Purpose of the Diagram

The UML class diagram explains how the Vaultique system is structured and how the different classes interact with each other. It also demonstrates how digital memories are connected to physical storage through QR code integration.

3.2 UML Class Diagram





Section 4



Task distribution:

Task	Student names
Project organization	Safaa Ahmed Sobh
Test Scenarios	Safaa Ahmed Sobh, Layan Alshreef, Nour Zehrawi
Test Cases	Safaa Ahmed Sobh, Layan Alshreef, Nour Zehrawi
Bug Report	Ethar Hatem Tamim



Test Scenarios:

- Scenario 1:

TS01: QR Code Scanning

The system should allow the user to scan a QR code and open the related digital chapter correctly.

- Scenario 2:

TS02: Collaborative Memory Management

Ensure the system allows multiple users to contribute to, edit, and update shared digital memories or chapters in real-time, maintaining data consistency across all devices.

- Scenario 3:

TS03: User Login

The system should allow registered users to access their accounts using valid credentials.



Test Cases:

- Test Case 1:

Test Case ID: TC01

Related Scenario: TS01

Description: Verify that the user can scan a QR code correctly.

Steps:

1. Open the Vaultique application.
2. Open the QR scanner.
3. Scan a valid QR code.
4. Wait for the system response.

Expected Result: The system should open the correct digital chapter.

Actual Result: The correct digital chapter was displayed successfully.

Status: Pass

- Test Case 2:

Test Case ID: TC02

Related Scenario: TS02

Description: Verify that a user can successfully add a collaborator to a shared memory chapter.

Steps:

1. Open the Vaultique mobile application.
2. Navigate to the "Create a memory" screen and select "**Collaborative Memory**".
3. Enter a "Memory Title".
4. In the "Add Collaborators" section, enter a valid username or email and click "**Add**".
5. Wait for the system to process the invitation.

Expected Result: The system should add the specified collaborator and show them in the contributors list for that chapter.

Actual Result: The collaborator was successfully invited and granted shared access.

Status: Pass.



- **Test Case 3:**

Test Case ID: TC03

Related Scenario: TS03

Description: Verify that a registered user can log in to the Vaultique application successfully.

Steps:

1. Open the Vaultique application.
2. Navigate to the Login page.
3. Enter a valid email and the correct password.
4. Click on the "Login" button.

Expected Result: The system should authenticate the user and redirect them to the home page.

Actual Result: The user was successfully authenticated and redirected to the home page.

Status: Pass



Bug Report:

Bug ID: B01

Title: Newly added "Memories" are not reflected on the "Chapters" page after clicking the save button.

Description: After adding the memories and getting the confirmation pop-up, the list of "Chapters" does not refresh to display the new record. The user will perceive a failure and must exit and re-enter the application to refresh the data.

Reproduce Steps:

1. Go to one of the "Chapters" tabs, press the add button (+).
2. Select the picture, fill in some details regarding the picture, and click the "Save" button.
3. Check that there are no changes reflected in the list of memories shown once going back to that particular "Chapter" tab.

Expected Result: The UI updates to reflect the new addition at the top of the list.

Actual Result: No changes appear in the list; only old data is shown until the user restarts the application.

Status: Open

Bug ID: B02

Title: Inability to display the "Memory Date" in a readable format on the display screens.

Description: The date is not displayed in an understandable technical format (2026-05-10T14:30:00Z) but rather in a simpler format, influencing the design aspect of the interface.

Reproduce Steps:

1. Start the Vaultique application and add a memory.
2. Pick a date by using the Date Picker function and press Save.
3. Open the "Chapters" tab or the details page of the selected memory.
4. View the format used for the date displayed at the top of the page.

Expected Result: The date is to be displayed in a clear format (10 May 2026).

Actual Result: The date is displayed in a purely technical format (Raw String), obtained straight from the database, which causes overlapping text problems.

Status: Open



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